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Overcoming Metaphysics: Carnap and Heidegger

It is well known that Rudolf Carnap (in Carnap 1932e) uses examples from Martin Heidegger as illustrations of metaphysical pseudo-sentences — including, most famously, the sentence "Nothingness itself nothings [*Das Nichts selbst nichtet*]" (Heidegger 1929b). It is tempting today for those on both sides — for those who sympathize with Carnap and those who sympathize with Heidegger alike — to view this episode with a more or less tolerant smile. Among those sympathetic to Carnap, Heidegger's sentence now appears as simply unintelligible, but hardly dangerous, nonsense: one is by no means surprised by such obvious absurdities coming from a fuzzy-minded "continental" thinker. Among those sympathetic to Heidegger, Carnap's criticism now appears as a case of simple blindness to Heidegger's point: one cannot expect a narrow-minded "analytic" philosopher even to begin to grasp such profundities. What both sides miss, I believe, is the depth and force this encounter had for Carnap and Heidegger themselves. We thereby miss the meaning and extent of the common context within which both contemporary philosophical traditions — both "continental" and "analytic" traditions — arise and develop.

The first point to notice is that Carnap and Heidegger had earlier met one another: at a celebrated disputation, or *Arbeitsgemeinschaft,* between Heidegger and Ernst Cassirer that took place during the International University Course held at Davos, Switzerland, from 17 March through 6 April 1929.[1] It was on this occasion, two years after the sensational appearance of *Being and Time,* that Heidegger first made public a radical phenomenological-metaphysical interpretation of the *Critique of Pure Reason* developed in explicit opposition to the Marburg school of neo-Kantianism with which Cassirer was closely associated — an interpretation Heidegger then wrote up in a few short weeks following the Davos university course and published as *Kant and the Problem of Metaphysics.*[2] Heidegger thereby presented himself — with extraordinary success, as it turned out — as the author of a fundamentally new kind of philosophy destined to replace the hegemony of the neo-Kantian tradition and to supplant the remaining "rationalist" tendencies in Husserlian phenomenology as well. In July 1929 Heidegger symbolically completed this ascension when he delivered his inaugural address as Edmund Husserl's successor to the chair of philosophy at Freiburg: Heidegger 1929b is the published record of this address. Carnap attended the Davos university course and reported on the occasion in his diary.[3] Like everyone else in attendance he appears to have been especially caught up in the intellectual excitement of the encounter between Heidegger and Cassirer. Moreover, Carnap was clearly impressed by Heidegger and had several philosophical conservations with him (ASP 025-73-03, entries from 18 March, 30 March, and 3 April 1929). When Carnap returned to Vienna he retained this sense of excitement and seems, in fact, to have studied *Being and Time* rather seriously. In particular, he actively participated in a discussion group in the summer of 1930 led by Heinrich Gomperz and Karl Bühler where Heidegger's book was intensively examined.[4] The first draft of Carnap 1932e was then written up in November 1930. Carnap presented it as lectures at Warsaw (November 1930), Zurich (January 1931), and Prague (November 1931) and then (in a revised version) at Berlin (July 1932) and Brünn (December 1932).[5] The published version appeared in *Erkenntnis,* the official journal of the Vienna Circle, in 1932. In §5 of Carnap 1932e, entitled "Metaphysical Pseudo-Sentences," Carnap introduces his consideration of examples from Heidegger 1929b by remarking that, although he "could just as well have selected passages from any other of the numerous metaphysicians of the present or the past," he has here chosen to "select a few sentences from that metaphysical doctrine which at present exerts the strongest influence in Germany."

The second point to notice is that Carnap's analysis and criticism of "Nothingness itself nothings" is more sophisticated and penetrating than one might have antecedently expected. For, on the one hand, Carnap's complaint is not that the sentence in question is unverifiable in terms of sense-data; nor is the most important problem that the sentence coins a bizarre new word and thus violates ordinary usage. The main problem is rather a violation of the logical form of the concept of nothing. Heidegger uses the concept both as a substantive and as a verb, whereas modern logic has shown that it is neither: the logical form of the concept of nothing is constituted solely by existential quantification and negation. On the other hand, however, Carnap also clearly recognizes that this kind of criticism would not affect Heidegger himself in the slightest; for the real issue between the two lies in the circumstance that Heidegger denies while Carnap affirms the philosophical centrality of logic and the exact sciences. Carnap accordingly refers to such Heideggerian passages as the following:

> [N]othingness is the source of negation, not vice versa. If the power of the understanding in the field of questions concerning nothingness and being is thus broken, then the fate of the dominion of "logic" within philosophy is also decided therewith. The idea of "logic" itself dissolves in a vortex of more original questioning.
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> The supposed soberness and superiority of science becomes ridiculous if it does not take nothingness seriously. Only because nothingness is manifest can science make what exists itself into an object of investigation. Only if science takes its existence from metaphysics can it always reclaim anew its essential task, which does not consist in the accumulation and ordering of objects of acquaintance but in the ever to be newly accomplished disclosure of the entire expanse of truth of nature and history.
>
> Therefore no rigor of a science can attain the seriousness of metaphysics. Philosophy can never be measured by the standard of the idea of science.[6]

Carnap concludes, in his own characteristically sober fashion: "We thus find a good confirmation for our thesis; a metaphysician here arrives himself at the statement that his questions and answers are not consistent with logic and the scientific mode of thinking" (Carnap 1932e, 232 [72]).

Heidegger's "Postscript" to Heidegger 1929b — published in the fourth edition in 1943 — considers three types of criticism that have been directed at the original lecture. Heidegger reserves his most extensive and militant response for the third criticism: namely, that "the lecture decides against 'logic.' " The heart of his response is as follows:

> The suspicion directed against "logic," whose conclusive degeneration may be seen in logistic [that is, modern mathematical logic], arises from the knowledge of that thinking that finds its source in the truth of being, but not in the consideration of the objectivity [*Gegenständlichkeit*] of what exists. Exact thinking is never the most rigorous thinking, if rigor [*Strenge*] receives its essence otherwise from the mode of strenuousness [*Anstrengung*] with which knowledge always maintains the relation to what is essential in what exists. Exact thinking ties itself down solely in calculation with what exists and serves this [end] exclusively. (Heidegger 1943, 104 [356])

It is clear, then, that Heidegger and Carnap are actually in remarkable agreement. "Metaphysical" thought of the type Heidegger is trying to awaken is possible only on the basis of a prior overthrow of the authority and primacy of logic and the exact sciences. The difference is that Heidegger eagerly embraces such an overthrow, whereas Carnap is determined to resist it at all costs.

The above sheds considerable light, I believe, on the context and force of Carnap's antimetaphysical attitude. For, by rejecting "metaphysics" as a field of cognitively meaningless pseudosentences, Carnap is by no means similarly rejecting all forms of traditional philosophy. He makes this perfectly clear, in fact, in his "Remarks by the Author" appended to the English translation of Carnap 1932e in 1957:

> *To section 1, "metaphysics."* This term is used in this paper, as usually in Europe, for the field of alleged knowledge of the essence of things which transcends the realm of empirically founded, inductive science. Metaphysics in this sense includes systems like those of Fichte, Schelling, Hegel, Bergson, Heidegger. But it does not include endeavors toward a synthesis and generalization of the results of the various sciences. (Carnap 1932e, [80])

In Carnap's reply to Paul Henle in Schilpp 1963 the point is made even more explicitly:

> Note that the characterization as pseudo-statements does not refer to all systems or theses in the field of metaphysics. At the time of the Vienna Circle, the characterization was applied mainly to those metaphysical systems which had exerted the greatest influence upon continental philosophy during the last century, viz., the post-Kantian systems of German idealism and, among contemporary ones, those of Bergson and Heidegger. On the basis of later, more cautious analyses, the judgment was not applied to the main theses of those philosophers whose thinking had been in close contact with the science of their times, as in the cases of Aristotle and Kant; the latter's epistemological theses about the synthetic a priori character of certain judgments were regarded by us as false, not as meaningless.[7]

So Carnap is primarily concerned with "overcoming" a very particular kind of "metaphysics": the main target is the post-Kantian German idealism he views as dominating recent European thought, and he views Heidegger, in particular, as the contemporary embodiment of such metaphysical dominance.

When Carnap emigrated to the United States in December 1935, he was therefore especially relieved to have finally left this European metaphysical tradition behind:

> I was not only relieved to escape the stifling political and cultural atmosphere and the danger of war in Europe, but was also very gratified to see that in the United States there was a considerable interest, especially among the younger philosophers, in the scientific method of philosophy, based on modern logic, and that this interest was growing from year to year.
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> In 1936, when I came to this country, the traditional schools of philosophy did not have nearly the same influence as on the European continent. The movement of German idealism, in particular Hegelianism, which had earlier been quite influential in the United States, had by then almost completely disappeared. Neo-Kantian philosophical conceptions were represented here and there, not in an orthodox form but rather influenced by recent developments in scientific thinking, much like the conceptions of Cassirer in Germany. Phenomenology had a number of adherents mostly in a liberalized form, not in Husserl's orthodox form, and even less in Heidegger's version. (Carnap 1963a, 34, 40)

Carnap's sense of liberation in thus escaping the "stifling" political, cultural, and philosophical atmosphere in Central Europe is palpable.

It is important, then, to understand Carnap's antimetaphysical attitude in its philosophical, cultural, and political context. Carnap's concern for this broader context is characteristic of him and, in fact, formed one of the main bonds uniting him with his more activist friend and colleague Otto Neurath. Neurath himself, as is well known, contributed an extremely engaged, neo-Marxist perspective to the Vienna Circle. Indeed, he had served as economics minister in Ernst Toller's short-lived Bavarian Soviet Republic in 1919 and received an eighteen-month sentence when it was crushed. As is also well known, an especially striking example of Carnap's own attitude toward the relationship between the philosophical work of the Vienna Circle and this wider cultural and political context is the preface to the *Aufbau,* dated May 1928. After calling for a radically new scientific, rational, and anti-individualistic conception of philosophy that is to emulate the slow process of mutual cooperation and collaboration typical of the special sciences, Carnap continues:

> We cannot hide from ourselves the fact that trends from philosophical-metaphysical and from religious spheres, which protect themselves against this kind of orientation, again exert a strong influence precisely

at the present time. Where do we derive the confidence, in spite of this, that our call for clarity, for a science that is free from metaphysics, will prevail? — From the knowledge, or, to put it more cautiously, from the belief, that these opposing powers belong to the past. We sense an inner kinship between the attitude on which our philosophical work is based and the spiritual attitude that currently manifests itself in entirely different spheres of life. We sense this attitude in trends in art, especially in architecture, and in the movements that concern themselves with a meaningful structuring [*Gestaltung*] of human life: of personal life and the life of the community, of education, of external organization at large. We sense here everywhere the same basic attitude, the same style of thinking and working. It is the orientation that is directed everywhere towards clarity yet recognizes at the same time the never entirely comprehensible interweaving of life, towards care in the individual details and equally towards the greater shape of the whole, towards the bonds between men and equally towards the free development of the individual. The belief that this orientation belongs to the future inspires our work. (Carnap 1928a, x–xi [xvii–xviii])

And, as Carnap explains in his diary, he is here expressing precisely the attitude that he and Neurath share.[8]

Carnap suggests that his and Neurath's orientation has much in common with that of modern architecture and, in particular, with that of the Dessau Bauhaus — a point that is borne out by the recollections of Herbert Feigl:

Carnap and Neurath also had a great deal in common in that they were somewhat utopian social reformers — Neurath quite actively, Carnap more "philosophically." . . . I owe [Neurath] a special debt of gratitude for sending me (I think as the first "emissary" of the Vienna Circle) to Bauhaus Dessau, then, in 1929, a highly progressive school of art and architecture. It was there in a week's sojourn of lectures and discussions that I became acquainted with Kandinsky and Klee. Neurath and Carnap felt that the Circle's philosophy was an expression of the *neue Sachlichkeit* which was part of the ideology of the Bauhaus. (Feigl 1969, 637)

Carnap's basic philosophical-political orientation is thus best expressed by the *neue Sachlichkeit* (the new objectivity, soberness, matter-of-factness): a social, cultural, and artistic movement committed to internationalism, to some form of socialism,[9] and, above all, to a more objective, scientific, and anti-individualistic reorganization of both art and public life inspired equally by the new Russian communism and the new American technology.[10]

Carnap and Heidegger are therefore at opposite ends of the spectrum not only philosophically but also in cultural and political terms.[11] And I think there is no doubt that this cultural and political dimension of their disagreement represents at least part of the explanation for Carnap's choosing Heidegger for his examples of metaphysical pseudosentences in Carnap 1932e. Indeed, particularly when read in the context of such programmatic statements as the preface to the *Aufbau,* this is already suggested by the sentence from §5, quoted above, where Carnap explains that he has here chosen to "select a few sentences from that metaphysical doctrine which at present exerts the strongest influence in Germany." Such a wider cultural and political context is also suggested by a passage in §6, where Carnap explains that the method of logical analysis has both a negative aspect (antimetaphysical) and a positive aspect (constructive analysis of science): "This negative application of method is necessary and important in the present historical situation. But the positive application — even already in contemporary practice — is more fruitful."[12] Carnap expresses this last idea in stronger and more militant terms, however, in the second version of his paper, the one presented at the lectures in Berlin and Brünn in July and December 1932. In this version the lecture closes with a discussion of the positive task of the method of logical analysis (that is, the clarification of the sentences of science) and, in particular, with the following words:

> These indications [are presented] only so that one will *not* think that the *struggle* [*Kampf*] *against metaphysics is our primary task.* On the contrary: in the meaningful realm [there are] many tasks and difficulties, there will always be enough struggle<?>. The struggle against metaphysics is only necessary because of the historical situation, in order to reject hindrances. There will, I hope, come the time when one no longer needs to present lectures against metaphysics. (ASP 110-07-19, p. 4)

One can imagine that this statement, coming at the very end of Carnap's lecture at Berlin in July 1932, had a much more dramatic impact than the more subtle suggestions buried in the published version.[13]

Neurath, for his part, dispenses entirely with all such subtleties. He never tires, for example, of characterizing "metaphysicians" and "school philosophers" — among whom Heidegger is a prominent representative — as enemies of the proletariat:

> Science and art are today above all in the hands of the ruling classes and will also be used as instruments in the class struggle against

> the proletariat. Only a small number of scholars and artists place themselves on the side of the coming order and set themselves up as protection against this form of reactionary thought.
>
> The idealistic school philosophers of our day from Spann to Heidegger want to rule, as the theologians once ruled; but the scholastics could support themselves on the substructure of the feudal order of production, whereas our school philosophers do not notice that their substructure is being pulled out from beneath their feet.[14]

Neurath was particularly ill-disposed toward attempts by such thinkers as Heinrich Rickert, Wilhelm Dilthey, and Heidegger to underwrite a special status for the *Geisteswissenschaften* in relation to the *Naturwissenschaften* (the humanities in relation to the natural sciences) — which attempts, according to Neurath, constitute one of the principal obstacles to rational and scientific social progress.[15]

That Carnap was in fact in basic agreement with Neurath here emerges clearly in a conversation he records after his final lecture presentation of Carnap 1932e at Brünn in December 1932:

> My lecture "Die Überwindung der Metaphysik" (Il ... had added: and the world-view of modern philosophy) in the banquet hall. Pretty well attended, lively participation, 1¼ hours. Afterwards various questions. *Erkenntnis* is here completely unknown. Then to a cafe. Prof. B. . . . , chemist, gives philosophical courses at the popular university, will report on my lecture in the socialistic ... newspaper. Marxist, is pleased with my Marxist views on how metaphysics will be overcome through reformation [*Umgestaltung*] of the substructure. (ASP 025-73-03, entry for 10 December 1932)

There can be very little doubt, therefore, that Carnap's attack on Heidegger, articulated and presented at an extraordinarily critical moment during the last years of the Weimar Republic, had more than purely philosophical motivations — or, perhaps better: that Carnap, like Neurath, conceived his philosophical work (and the attack on Heidegger in particular) as a necessary piece of a much larger social, political, and cultural struggle.[16]

It is noteworthy, finally, that Heidegger was aware of Carnap's attack and, indeed, explicitly responded to it: in a part of his 1935 lecture course "Introduction to Metaphysics" that does not appear in the published version in 1953.[17] Heidegger explains how, with the collapse of German idealism in the second half of the nineteenth century, the philosophical understanding of Being degenerated into a consideration of the "is" — that is, a logical consideration of the propositional copula. He continues in a memorable paragraph that is worth quoting in full:

> Going further in this direction, which in a certain sense has been marked out since Aristotle, and which determines "Being" from the "is" of the proposition and thus finally destroys it, is a tendency of thought that has been assembled in the journal *Erkenntnis.* Here the traditional logic is to be for the first time grounded with scientific rigor through mathematics and the mathematical calculus, in order to construct a "logically correct" language in which the propositions of metaphysics — which are all pseudo-propositions — are to become impossible in the future. Thus, an article in this journal (2:1931–32, 219ff.) bears the title "Überwindung der Metaphysik durch logische Analyse der Sprache." Here the most extreme flattening out and uprooting of the traditional theory of judgment is accomplished under the semblance of mathematical science. Here the last consequences of a mode of thinking which began with Descartes are brought to a conclusion: a mode of thinking according to which truth is no longer disclosedness of what exists and thus accommodation and grounding of Dasein in the disclosing being, but truth is rather diverted into *certainty* — to the mere securing of thought, and in fact the securing of mathematical thought against all that is not thinkable by it. The conception of truth as the securing of thought led to the definitive profaning [*Entgötterung*] of the world. The supposed "philosophical" tendency of mathematical-physical positivism wishes to supply the grounding of this position. It is no accident that this kind of "philosophy" wishes to supply the foundations of modern physics, in which all relations to nature are in fact destroyed. It is also no accident that this kind of "philosophy" stands in internal and external connection with Russian communism. And it is no accident, moreover, that this kind of thinking celebrates its triumph in America. All of this is only the ultimate consequence of an apparently merely grammatical affair, according to which Being is conceived through the "is," and the "is" is interpreted in accordance with one's conception of the proposition and of thought. (Heidegger 1983, 227–28)[18]

Thus Heidegger, in terms no more subtle than Neurath's, once again expresses a rather remarkable agreement with Carnap concerning the underlying sources of their opposition — which, as is now clear, extend far beyond the purely philosophical issues between them.

These philosophical issues, as we have seen, are based in the end on a stark and profound disagreement about the nature and centrality of logic. Thus Carnap criticizes "Nothingness itself nothings" primarily on the grounds of logical form: modern mathematical logic shows that the concept of nothing is to be explained in terms of existential quantification and negation and can therefore by no means function either as a substantive (individual constant) or as a verb (predicate). For Heidegger, by contrast, such a purely logical analysis misses precisely his point: what he calls nothingness is prior to logic and hence prior, in particular, to the concept of negation. In tracing out the roots of this fundamental disagreement over the philosophical centrality of logic, it turns out, we need to appreciate the extent to which the thought of both philosophers arises from the neo-Kantian tradition that dominated the German-speaking world at the end of the nineteenth and beginning of the twentieth century — a tradition within which both men received their philosophical training.

There were in fact two quite distinguishable versions of neo-Kantianism that were dominant at the time: the so-called Marburg school of neo-Kantianism founded by Hermann Cohen and then continued by Paul Natorp and (at least until about 1920) Ernst Cassirer, and the so-called Southwest school of neo-Kantianism founded by Wilhelm Windelband and systematically developed by Heinrich Rickert. The former school emphasized the importance of mathematics and natural science and, in fact, saw the true achievement of the *Critique of Pure Reason* as a laying of the groundwork for Newtonian mathematical physics. The latter school, by contrast, emphasized the distinctive importance of the *Geisteswissenschaften* and, accordingly, devoted considerable philosophical efforts to articulating a sharp methodological distinction between the latter and the *Naturwissenschaften.* Heidegger studied with Rickert at Freiburg (before Rickert succeeded Windelband at Heidelberg) — completing his habilitation under Rickert in 1915. Carnap, for his part, studied Kant at Jena with Bruno Bauch — another student of Rickert's from Freiburg — and, in fact, completed his doctoral dissertation under Bauch in 1921.[19] It is clear, moreover, that Carnap carefully studied both versions of neo-Kantianism and, in particular, the writings of Natorp, Cassirer, and Rickert.[20]

Common to both versions of neo-Kantianism is a certain conception of epistemology and the object of knowledge inherited from Kant.[21] Our knowledge or true judgments should not be construed, according to this conception, as representing or picturing objects or entities that exist independently of our judgments — whether these independent entities are the "transcendent" objects of the metaphysical realist existing somehow "behind" our sense-experience or the naked, unconceptualized sense-experience itself beloved of the empiricist. In the first case ("transcendent" objects), knowledge or true judgment would be impossible for us, since, by hypothesis, we have absolutely no independent access to such entities by which we could verify whether the desired relation of representation or picturing holds. In the second case (naive empiricism), knowledge or true judgment would be equally impossible, however, for the stream of unconceptualized sense-experience is in fact utterly chaotic and intrinsically undifferentiated: comparing the articulated structures of our judgments to this chaos of sensations simply makes no sense. How, then, is knowledge or true judgment possible? What does it mean for our judgments to relate to an object? The answer is given by Kant's "Copernican Revolution": the object of knowledge does not exist independently of our judgments at all; on the contrary, this object is first created or "constituted" when the unconceptualized data of sense are organized or framed within the a priori logical structures of judgment itself. In this way, the initially unconceptualized data of sense are brought under a priori "categories" and thus first become capable of empirical objectivity.

Yet there is a crucially important difference between this neo-Kantian account of the object of knowledge and judgment and Kant's original account. For Kant, we cannot explain, on the basis of the a priori logical structures of judgment alone, how the object of knowledge becomes possible. We need additional a priori structures that mediate between the pure forms of judgment comprising what Kant calls general logic and the unconceptualized manifold of impressions supplied by the senses: these mediating structures are the pure forms of sensible intuition — space and time. Thus the pure logical forms of judgment only become categories when they are "schematized," that is, when they are given a determinate spatio-temporal content in relation to the pure forms of sensible intuition. The pure logical form of a categorical judgment, for example, becomes the category of *substance* when it is schematized in terms of the temporal representation of permanence; the pure logical form of a hypothetical judgment becomes the category of *causality* when it is schematized in terms of the temporal representation of succession; and so on. For Kant, then, pure formal logic (general logic) must, if it is to play an epistemological role, be supplemented by what he calls transcendental logic: with the theory of how logical forms become schematized in terms of pure spatio-temporal representations belonging to the independent faculty of pure intuition. And it is precisely this theory, in fact, that forms the heart of the transcendental analytic of the

Critique of Pure Reason: the so-called metaphysical and transcendental deductions of the categories.

Now both versions of neo-Kantianism entirely reject the idea of an independent faculty of pure intuition. The neo-Kantians here follow the tradition of post-Kantian idealism in vigorously opposing the dualistic conception of mind characteristic of Kant's own position: the dualism, that is, between a logical, conceptual, or discursive faculty of pure understanding and an intuitive, nonconceptual, or receptive faculty of pure sensibility. For the neo-Kantians, the a priori formal structures in virtue of which the object of knowledge becomes possible must therefore derive from the logical faculty of the understanding and from this faculty alone. And, in this way, epistemology or "transcendental logic" becomes the study of purely logical, purely conceptual, *and thus essentially non–spatio-temporal* a priori structures. Space and time, conceived as Kantian pure forms of sensible intuition, can no longer play a role in our explanation of how the object of knowledge and judgment becomes possible.

It is this last feature of their conception of epistemology, moreover, that associates the neo-Kantians (again in both versions) with Husserlian phenomenology and, in particular, with the polemic against psychologism of Husserl's *Logical Investigations.* For the neo-Kantians had also arrived — albeit by a different route — at a conception of pure thought or pure logic whose subject matter is an essentially nontemporal, and therefore certainly not psychological, realm: an "ideal" realm of timeless, formal-logical structures. Indeed, Husserl's conception of "pure logic" — which was generally recognized to have its sources in the earlier work of Bernhard Bolzano, Johann Herbart, Rudolf Lotze, and Alexius Meinong — can be fairly characterized as the dominant idea of the period. Accordingly, it plays a central role not only in the thought of the neo-Kantians of the Marburg and Southwest traditions but also in the early thought of both Carnap and Heidegger.[22]

We shall have to return to the relationship between Husserl and Heidegger shortly. But it is first necessary to appreciate the fundamental problems facing the epistemology of the neo-Kantians (in both versions) — problems that flow directly from their enthusiastic embrace of the idea of "pure logic." For, as we have just seen, the logical forms of judgment, in Kant's original conception, become categories — and thus make the object of knowledge possible — precisely through their prior application to the pure forms of sensible intuition. It is on this basis, and on this basis alone, that we can explain how the purely analytic forms of thought apply to the spatio-temporal world of sense so as to make synthetic knowledge of empirical objects (that is, of appearances) possible. Yet the neo-Kantians entirely reject the idea of such an intermediate faculty of pure intuition and hence the central Kantian conception of the schematism of the pure concepts of the understanding as well. How, then, are the pure forms of thought — now conceived as belonging wholly to an "ideal," essentially non–spatio-temporal realm — supposed to apply to the spatio-temporal world of sense? How do the categories make the object of (empirical) knowledge possible?

It is in attempting to answer these questions that the two schools of neo-Kantian epistemology strikingly diverge from one another. The Marburg school, as indicated above, continues to follow Kant in taking mathematical physics as the paradigm of objective knowledge. That school's most basic move, accordingly, is to "mathematize" the pure forms of thought. Beginning with Cohen's attempt to assimilate the fundamental moment of judgment to the mathematical concept of the differential, this line of thought reaches its culmination in Cassirer's *Substance and Function* — where logic is identified with the pure theory of relations developed especially (building on the work of David Hilbert, Georg Cantor, and Richard Dedekind) in Bertrand Russell's *The Principles of Mathematics.* The timeless realm of "pure thought" is thus identified with the totality of what we now call relational structures — the "objects" thereof being simply abstract "places" within such a relational structure. In empirical knowledge, on the other hand, we develop an essentially nonterminating — but in some sense converging — *sequence* of relational structures, each element of which represents the state of mathematical physics at some particular point in the methodological history of science. (That this sequence does not terminate thus constitutes the essential point of difference between pure and empirical knowledge.) The object of empirical knowledge — the sensible world — is then conceived simply as the ideal limit or infinitely distant *X* toward which the methodological sequence of science is converging.[23] The Marburg school thereby solves the problem of the categories by a kind of "logicization" of the object of empirical knowledge, and it is with good reason, then, that the school's epistemological conception becomes known as "logical idealism."

Within the Southwest school, on the other hand, logic and the realm of "pure thought" are sharply and explicitly separated from mathematics. And this fundamental divergence between the two schools emerges with particular clarity in a dispute between Natorp and Rickert in the years 1910–11. Natorp argues that the concept of number belongs to "pure thought" and thus neither to pure intuition nor to psychology (Natorp 1919). Rickert directly challenges Natorp's conception — arguing that the concept of one as a number (as the first element of the number series) cannot be derived from the logical concepts of identity and difference and is therefore "alogical" (Rickert 1911). For Rickert, the numerical concept of one (unlike the logical concepts of identity and difference) does not apply to all objects of thought as such but presupposes that we are antecedently given objects arranged in a homogeneous serial order. The numerical concept of quantity can therefore not belong to logic — where logic is here being clearly identified with traditional syllogistic logic.[24] In this approach, since we neither follow Kant in invoking a mediating faculty of pure intuition nor follow the Marburg school in "logicizing" the object of empirical knowledge after the example of mathematical physics, we are therefore left only with the forms of judgment of traditional logic on the one side and the spatio-temporal manifold of given empirical objects on the other. So, as one would expect, particularly vexing problems in attempting to explain the application of the former to the latter arise within the Southwest school.

The underlying tensions expressed in the epistemology of the Southwest school become painfully evident in the work of Emil Lask, a brilliant student of Rickert's who then held an associate professorship at Heidelberg and was killed in the Great War in 1915. The basic argument of Lask 1912 is that whereas the Kantian philosophy has indeed closed the gap between knowledge and its object, we are nonetheless left with a new gap between what Lask calls "transcendental," "epistemological," or "material" logic, on the one side, and "formal" logic, on the other. Formal logic is the subject matter of the theory of judgment — the realm of necessarily valid and timeless "senses," "objective thoughts," or "propositions in themselves" familiar within the tradition of "pure logic."[25] Transcendental or material logic, on the other hand, is the theory of the categories in Kant's sense: the theory of how the concrete object of knowledge and experience is made possible by the activity of thought. But, and here is the central idea of Lask's argument, transcendental or material logic is *not* based on formal logic, and, accordingly, we explicitly reject Kant's metaphysical deduction — the entire point of which, as indicated above, is precisely to derive the categories from the logical forms of judgment.[26] For Lask, what is fundamental is the concrete, already categorized real object of experience: the subject matter of formal logic (comprising the structures of the traditional logical theory of judgment) only arises subsequently in an artificial process of abstraction, by which the originally unitary categorized object is broken down into form and matter, subject and predicate, and so on. Moreover, since this comes about due to a fundamental weakness or peculiarity of our human understanding — our inability to grasp the unitary categorized object *as* a unity — all the structures of "pure logic," despite their undoubtedly timeless and necessary status, are in the end artifacts of subjectivity. Since the pure forms of judgment of traditional logic are now seen as entirely bereft of the capacity even to begin the "constitution" of any real empirical object, the entire realm of "pure logic" appears as nothing but an artificially constructed intermediary possessing no explanatory power whatsoever.[27]

Heidegger's earliest philosophical works, as noted above, fall squarely within the antipsychologistic tradition of "pure logic" and receive their primary orientation (not surprisingly) from his teacher Rickert. Accordingly, these early investigations revolve around the central distinctions between psychological act and logical content, between real thought process and ideal atemporal "sense," between being (*Sein*) and validity (*Geltung*). For, as Lotze in particular has shown, the realm of the logical has a completely different mode of existence (validity or *Geltung*) than that of the realm of actual spatio-temporal entities (being or *Sein*).[28] Moreover, as Rickert has shown, the realm of the logical (the realm of validity) is also distinct from that of the mathematical; for, although the latter is equally atemporal and hence equally ideal, it presupposes a particular object — the existence of "quantity" — and therefore lacks the complete generality characteristic of the logical. It follows that we must sharply distinguish the realm of the logical both from the given heterogeneous qualitative continuum of empirical reality and from the homogeneous quantitative continuum of mathematics.[29] In emphasizing these fundamental distinctions and, above all, in maintaining "*the absolute primacy of valid sense* [*den absoluten Primat des geltenden Sinnes*],"[30] Heidegger shows himself to be a faithful follower of Rickert indeed.

Yet, as we have just seen, Rickert's fundamental distinctions lead naturally to fundamental problems — problems that stand out especially vividly in the work of Lask. In particular, once we have delimited the realm of the logical so sharply from all "neighboring" realms, it then becomes radically unclear how the realm of the logical is at all connected with the real world of temporal being [*Sein*]: with either the realm of empirical nature where the objects of our (empirical) cognition reside or the realm of psychological happenings where our acts of judgment reside. The realm of "valid sense," which was intended as an intermediary between these last two realms wherein our cognition of objects is "constituted" and thus made possible, thereby becomes deprived of all explanatory power. Now Heidegger, for his part, was of course most sensitive indeed to the difficult position in which Rickert had become entangled; and, as a consequence, he became increasingly attracted to the more radical position of Lask,[31] which Heidegger also saw as having essential ideas in common with the emerging new phenomenology being developed by Edmund Husserl. After Rickert left Freiburg to take Windelband's chair at Heidelberg and Husserl left Göttingen to take Rickert's chair at Freiburg in 1916, Heidegger became an enthusiastic proponent of the new phenomenology and, in particular, distanced himself further and further from Rickert. This distance from Rickert became quite extreme by 1925–26, when Heidegger was completing the work on *Being and Time,* and it is graphically evident in lectures Heidegger presented at Marburg in the summer semester of 1925 and the winter semester of 1926. Here Heidegger speaks of Rickert with almost undisguised contempt, whereas Husserl appears as the leader of a new "breakthrough" in philosophy — a "breakthrough" that has decisively overcome neo-Kantianism.[32]

The first element of this "breakthrough" is a "direct realist" conception of truth as "identification" — a conception that can be seen as definitively rejecting the idea that formal logic is foundational for truth in general and thus as also overturning the "Copernican Revolution." For, according to the theory of truth articulated in volume 2 of the *Logical Investigations,* truth in general is not even propositional: it consists simply in the circumstance that an intention or meaning (whether propositional or not) is directly "identified" — in immediate intuition — with the very thing that is intended or meant. Thus, truth in general need involve none of the structures (subject and predicate, ground and consequent, and so on) studied in traditional formal logic. On the contrary, such peculiarly logical structures only emerge subsequently in the very special circumstances of "categorial intuition," where specifically propositional intentions or meanings are intuitively grasped in their most abstract — and, as it were, secondary and derivative — formal features. In this sense, then, Husserl's "direct realist" conception of the relationship between logical form and truth in general parallels Lask's view of the artificiality and subjectivity of logical form: in neither case can formal logic be in any way foundational or explanatory for truth as "relation to an object."[33]

The second element of the Husserlian "breakthrough" is just the idea of phenomenology as such — an idea that also emerges in volume 2 of the *Logical Investigations* as that of an "epistemology of the logical." For it is this idea, and this idea alone, that first opens up the possibility of bridging the gulf between the logical and the psychological created by the polemic against psychologism of volume 1. The problem, however, is to explain how such an "epistemology of the logical" —

which is to investigate the *relationship* between psychological act and logical or "essential" content — can itself avoid collapsing into psychologism. The answer is disarmingly simple: phenomenology is not empirical psychology because it aims to elucidate the underlying a priori structures or "essences" of psychological phenomena (the "essences" of perception, recollection, imaginative representation, and so on). Our investigation proceeds by means of "essential analysis [*Wesensanalyse*]" and "essential intuition [*Wesenserschauung*]" — and therefore is, in particular, entirely independent of the actual instances of the psychological structures in question that may or may not exist in the real world.[34] We are interested, that is, only in the purely ideal or "essential" structures of psychological phenomena: in "pure consciousness."[35]

Yet, for this very reason, Husserl's conception of pure phenomenology and "pure consciousness" could not be fully satisfactory from Heidegger's point of view. For Heidegger's problem — arising so painfully and vividly within the neo-Kantian epistemology of the Southwest school — was precisely that of the application of abstract and ideal "valid senses" to concrete and real objects of cognition: the problem of the application of the categories in Kant's sense to actual spatio-temporal objects. And *this* problem, it is clear, cannot be solved within the framework of Husserlian "pure consciousness"; for the latter, as we have just seen, itself belongs to the purely ideal realm of "essences" and is thus entirely independent of the existence of any and all concrete instances (whether of actual states of consciousness or of its empirical objects). It is by no means surprising, therefore, that we already find rumblings in a new and quite un-Husserlian direction in the concluding chapter added to the published version of Heidegger's habilitation in 1916. Heidegger there suggests that a genuine unification of time and eternity — of change and absolute validity — can be effected only through the concept of "living spirit [*der lebendige Geist*]" construed as a concrete and essentially historical subject. The "subjective logic" sought for by Rickert and Husserl requires a more fundamental point of view according to which the subject is no mere "punctiform" cognitive subject but an actual concrete subject comprehending the entire fullness of its temporal-historical involvements. And such an investigation of the concrete historical subject must, according to Heidegger, be a "translogical" or "metaphysical" investigation.[36] Thus, Heidegger is here already beginning to come to terms with the historically oriented *Lebensphilosophie* of Wilhelm Dilthey — an influence that will prove decisive in *Being and Time*.[37]

It is of course in *Being and Time*, completed ten years later, that Heidegger finally works out the desired "subjective logic" with a *con-*

crete subject — the so-called existential analytic of Dasein. Heidegger's concrete subject — Dasein, the concrete living human being — is distinguished from the "pure consciousness" of Husserlian phenomenology in three fundamental respects. First, Dasein necessarily exists in a world: a world of concrete spatio-temporal objects existing independently of it, which it does not create and over which it has only very limited control, and into which — without its consent, as it were — it is "thrown." Indeed, for Heidegger, Dasein essentially *is* such "being-in-the-world." Second, Dasein's relationship to this world is first and foremost practical and pragmatic rather than epistemic and contemplatively intuitive. The items in Dasein's world therefore appear to it originally as practically "ready-to-hand [*Zuhanden*]" — as environmental items to be used in the service of particular concrete projects — rather than as merely "present-to-hand [*Vorhanden*]" for theoretical inspection and consideration. Indeed, for Heidegger, theoretical cognition of the merely "present-to-hand" is a *derivative* mode of Dasein: a particular "modification" of the more basic, essentially practical and pragmatic, mode of involvement with the "ready-to-hand."[38] Finally, Dasein is essentially a historical being — in an important sense it is *the* historical being. For the essence or "being" of Dasein is "care [*Sorge*]" — roughly, the above-described orientation toward its world from the point of view of the totality of its practical involvements and projects — and the "ontological meaning of care" is *temporality,* where temporality in this sense is essentially historical and thus to be sharply and explicitly distinguished from the uniform and featureless "time" of natural science.[39]

There is no doubt, then, that Heidegger's Dasein is much more concrete than Husserl's "pure consciousness" — in the sense that the former has more of the features of a real human being than does the latter. From the point of view of Husserlian phenomenology, however, an obvious dilemma for Heidegger arises at this point. For what can Heidegger's "existential analytic of Dasein" possibly mean from Husserl's perspective? Either Heidegger is trying to describe the concrete reality of empirical human beings in their concrete and empirical character, in which case his enterprise is simply a branch of empirical anthropology having no specifically philosophical interest whatsoever; or Heidegger is trying to elucidate the "essence" or nature of the concrete human being by means of an "essential analysis" of that nature, in which case Heidegger, too, must perform the "eidetic" reduction and abstract from all questions involving the real existence of the entities under consideration. Thus, either Heidegger falls prey to the charge of naturalism and psychologism or his "existential analytic of Dasein" is in the end no closer to actual concrete reality than is Husserl's phenomenology.

It is in response to this dilemma, I believe, that Heidegger's true philosophical radicalism emerges; for it is precisely here — in attempting to construct an a priori analysis of the real concrete subject — that Heidegger fundamentally changes the terms within which the entire tradition of "pure logic" was articulated and, in fact, introduces a fundamentally new existential dimension into this tradition. For Heidegger, the "existential analytic of Dasein" can by no means be assimilated to a description of the "essence" of Dasein in the traditional meaning of this term (where "essence" or "whatness" is contrasted with "existence" or "thatness") precisely because the distinguishing feature of Dasein — as opposed to all other entities encountered within the world — is that Dasein has no "essence" of this kind at all: *What* Dasein is can be determined only by Dasein's own free choice in the face of its fundamental possibility of "being-toward-death" — a choice that can be either "authentic" and "resolute" or "inauthentic" and fallen into the "They" of everyday public existence.[40] In either case, however, since Dasein's "essence" or "whatness" depends in the end on its own free choice — and is thus in no sense simply "given" as in the case of entities encountered within the world — Dasein's "essence" cannot be meaningfully separated from its "existence" at all. Indeed, from this point of view, Dasein's "essence" *is* "existence." The dilemma just raised from the point of view of Husserlian phenomenology therefore has no force whatever from Heidegger's own point of view: by replacing phenomenological "essential analysis" with what he calls "existential-ontological analysis," Heidegger has transcended the traditional distinction between "essence" and "existence" and opened up the paradoxical-sounding possibility of an a priori analysis of concrete existence itself.[41]

At the same time, Heidegger has thereby definitively transcended the problematic of the neo-Kantian tradition as well. This comes out most clearly in §44 of *Being and Time,* entitled "Dasein, Disclosedness, and Truth," where Heidegger explicitly rejects the "Copernican Revolution" and the associated idea that truth is to be understood in terms of "valid judgment" in favor of an apparently "direct realist" account in which truth is conceived as a kind of immediate "disclosedness [*Erschlossenheit*]" or "uncoveredness [*Entdeckt-sein*]" of a being within the world — an account that explicitly invokes Husserl's notion of "identification."[42] Heidegger's "direct realism" is very special, however, for Dasein's most fundamental relation to the world is not a cognitive relation at all. Indeed, Dasein's most fundamental relation to the world is one of either "authentic" or "inauthentic" existence — in which Dasein's own peculiar mode of being (that is, "being-in-the-world") is itself either disclosed or covered over.[43] Moreover, in the moment of an "authentic" decision, Dasein must choose among possibilities already given in the world—possibilities that must themselves be somehow already "present" and available in Dasein's historically given situation. Dasein must thus appropriate its "fate" and, in precisely this way, is essentially historical.[44]

What is therefore central, for Heidegger, is the circumstance that we do not start with a merely cognitive subject together with its "contents of consciousness," but rather with a living practical subject — a subject that is essentially temporally finite and hence necessarily engaged with its historically given environmental situation. Assertion or judgment then appears as a "derivative mode of interpretation" in which the "hermeneutical 'as' " of practical understanding of the "ready-to-hand" (where an item "ready-to-hand" is understood "as" suited for a given end or purpose) is transformed into the "apophantical 'as' " of theoretical understanding of the "present-to-hand" (where an item "present-to-hand" is understood "as" determined by a given predicate). If we forget this derivative character, however, all the misunderstandings prevalent in "the presently dominant theory of 'judgment' oriented around the phenomenon of 'validity [*Geltung*]' " then arise. We end up, in particular, with Lotze's distinction between "being" and "validity." Since we begin, within this latter tradition, with a "Cartesian" subject entirely enclosed within the psychical realm of its own representations, we cannot explain truth as a relation between these representations and an object existing independently of them. Truth can therefore only mean what is constant and unchangeable in the flux of representations and is thus understood as "form" or "essence" in a quasi-Platonic sense standing over and against the realm of flux and change. We thereby arrive at the distinction between "being" and "validity," "real" and "ideal." And this distinction, by the "Copernican" conception of the object of knowledge, is now equated with the distinction between subjective and objective. Finally, since "objectivity" is thus equated with "validity" in the sense of atemporal or eternal "ideal being," "objectivity" is also equated with necessary *intersubjectivity:* with "bindingness" for all subjects.[45]

For Heidegger himself, by contrast, truth is in no way to be equated with "objectivity" in the sense of necessary and universal intersubjectivity. On the contrary, this most basic idea of the Kantian "Copernican Revolution" is definitively rejected in favor of a "direct realist" conception of truth as direct "disclosedness" to Dasein in a particular and irreducibly historical environmental situation: all truth is ultimately both particular and historical.[46] Indeed, to think otherwise is to refuse to acknowledge the essential particularity of an "authentic" decision in the face of "being-toward-death" and to seek refuge instead in the public everydayness of the "They."[47] In the end, therefore, it is Heidegger's radical transformation of the neo-Kantian tradition within which he was trained that underwrites his equally radical rejection of the priority and centrality of logic: his claim that the traditional theory of judgment based on the "is" of predication is itself necessarily derivative from a properly philosophical point of view.

We observed above that Carnap also received his philosophical training within the neo-Kantian tradition and, in fact, that he completed his doctoral dissertation in 1921 under Rickert's student Bruno Bauch — a dissertation in which Kantian themes predominate.[48] It was in the years immediately after finishing his dissertation, in 1922–25, that Carnap undertook most of the work on the project that was eventually to issue in *Der logische Aufbau der Welt* (Carnap 1963a, 16–19). And it was on this basis, moreover, that Carnap caught the attention of the "Philosophical Circle" that had gathered around Moritz Schlick at the University of Vienna. Carnap had become acquainted with Schlick in the summer of 1924 and was invited to give lectures to Schlick's circle in the winter of 1925. These lectures on the *Aufbau* project — which was at the time entitled "Entwurf einer Konstitutionstheorie der Erkenntnisgegenstände" (Outline of a constitutional theory of the objects of cognition) — were extremely well received: Carnap returned to Vienna as assistant professor, with his "Entwurf" (then being eagerly read within Schlick's circle) serving as his habilitation.[49] A revised version was finally published in 1928 under the now familiar title.

The aim of the *Aufbau,* as recent scholarship has made increasingly clear, is by no means exclusively to represent the point of view of phenomenalistic or extreme empiricist "positivism." Indeed, Carnap himself explains the relationship between "positivism" and neo-Kantianism as follows:

> Cassirer ([*Substanzbegr.*] 292ff.) has shown that a science having the goal of determining the individual through lawful interconnections [*Gesetzseszusammenhänge*] without its individuality being lost must apply, not class ("species") concepts, but rather *relational concepts;* for the latter can lead to the formation of series and thereby to the establishing of order-systems. It hereby also results that relations are necessary as first posits, since one can in fact easily make the transition from relations to classes, whereas the contrary procedure is only possible in a very limited measure.

> The merit of having discovered the necessary basis of the constitutional system thereby belongs to two entirely different, and often mutually hostile, philosophical tendencies. *Positivism* has stressed that the sole *material* for cognition lies in the undigested [*unverarbeitet*] experiential *given;* here is to be sought the *basic elements* of the constitutional system. *Transcendental idealism,* however, especially the neo-Kantian tendency (Rickert, Cassirer, Bauch), has rightly emphasized that these elements do not suffice; *order-posits* [*Ordnungssetzungen*] must be added, our "basic relations."[50]

Carnap does not intend simply to supplant neo-Kantianism by "positivism" in the *Aufbau:* he hopes, on the contrary, to retain the insights of *both* views.[51]

As Carnap suggests, the influence of Cassirer's *Substance and Function* is especially important to the *Aufbau.* This is not surprising, for the agreement between the two conceptions actually extends far beyond the emphasis on the significance of the modern logical theory of relations stressed here. According to *Substance and Function,* as indicated above, the theory of knowledge consists of two parts. On the one hand, we have the theory of the *concept* (part 1), which, for Cassirer, is given by the totality of pure relational structures provided by the new logic. On the other hand, however, we have the theory of *reality* (part 2), in which pure relational structures are successively applied in the methodological progress of mathematical natural science in such a way that a never completed — but convergent — sequence results. Thus, whereas pure mathematics is given by the collection of all pure or abstract relational structures, applied mathematics (mathematical physics) is given by an infinite methodological series of such structures. And it is this methodological series of abstract structures that, for Cassirer (and for the Marburg school more generally), represents the empirical side of knowledge given by "sensation." The concrete empirical world of sense-perception is not a separate reality existing somehow outside of this methodological series: it is simply the fully determinate and complete "limit theory" toward which this series is converging.

Now Carnap, in the *Aufbau,* also represents empirical knowledge by a serial or stepwise methodological sequence. This sequence is intended to represent, not so much the historical series of mathematical-physical successor theories, but rather the epistemological progress of a single individual or cognitive subject — in which its knowledge extends from the initial subjective sensory data belonging to the *autopsychological* realm, through the world of public external objects constituting the *physical* realm, and finally to the intersubjective and cultural realities belonging to the *heteropsychological* realm. Carnap's methodological series is thus a "rational reconstruction" intended formally to represent the "actual process of cognition."[52] For Carnap, as for Cassirer, we thereby represent the empirical side of knowledge by a methodological sequence of formal structures. For Carnap, however, this is not a sequence of historically given mathematical-physical successor theories but *a sequence of levels or ranks in the hierarchy of logical types* of Russell's and Whitehead's *Principia Mathematica* — a sequence of levels *ordered by type-theoretic definitions*. Objects on any level (other than the first) are thus defined as classes of objects (or relations between objects) from the preceding level.[53]

The construction or "constitution of reality" begins with "elementary experiences" — holistic momentary cross-sections of the stream of experience — ordered by a (holistically conceived) "basic relation" of remembrance-of-part-similarity-in-some-arbitrary-respect. The main formal problem within the autopsychological realm is then to differentiate — on this initially entirely holistic basis — the particular sense qualities and sense modalities from one another. After grouping elementary experiences into classes (and classes of classes...) thereof via the one given basic relation and a complex procedure of "quasi-analysis," Carnap is in a position to define the *visual field* as the unique sense modality possessing exactly five dimensions (two of spatial location and three of color quality).[54] On this basis we can then define the "visual things" in the physical realm: after embedding the visual fields of our subject in a numerical space-time manifold ($\mathbf{R}^4$), we project colored points of these visual fields along "lines of sight" onto colored surfaces in such a way that principles of constancy and continuity are satisfied. And, in an analogous fashion, we can then define the "physical things" or objects of mathematical physics: we coordinate purely numerical "physical state magnitudes" with sensible qualities in accordance with the laws and methodological principles of the relevant science (e.g., the electro-dynamic theory of light and color).[55] Finally, we can constitute the heteropsychological realm by, first, constructing other subjects of experience analogous to the initial subject (that is, systems of elementary experiences coordinated to "other" human bodies) and, second, constructing an "intersubjective world" common to all such subjects through an abstraction (via an equivalence relation) from the resulting diversity in "points of view."[56]

In this way Carnap's "constitution of reality" achieves a "logicization" of experience or the sensible aspects of reality parallel to that of the Marburg school. For the entire point of Carnap's method of "purely structural definite descriptions" (like that of the visual field sketched above) is to individuate the objects in question in purely formal-logical terms, making no reference whatever to their intrinsic or ostensive phenomenal qualities. The constitutional system thereby demonstrates that objective — that is, intersubjectively communicable — knowledge is possible *despite* its necessary origin in purely subjective experience.[57] Carnap characterizes the resulting kinship between his constitutional system and the "logical idealism" of the Marburg school as follows:

> Constitutional theory and *transcendental idealism* agree in representing the following position: all objects of cognition are constituted (in idealistic language, are "generated in thought"); and, moreover, the constituted objects are only objects of cognition *qua* logical forms constructed in a determinate way. This holds ultimately also for the basic elements of the constitutional system. They are, to be sure, taken as basis as unanalyzed unities, but they are then furnished with various properties and analyzed into (quasi-)constituents (§116); first hereby, and thus also first as constituted objects, do they become objects of cognition properly speaking — and, indeed, objects of psychology.[58]

Indeed, there is one important respect in which Carnap's conception is even more radical than that of the Marburg school. Cassirer's *Substance and Function,* for example, retains an element of dualism between pure thought and empirical reality: the contrast between the pure relational structures of logic and mathematics, on the one hand, and the historical sequence of successor theories representing the methodological progress of empirical natural science, on the other. For Carnap, by contrast, empirical reality simply *is* a particular logical structure: a type-theoretic structure (representing the epistemological progress of an initial cognitive subject) erected on the basis of a single, primitive, nonlogical relation.[59]

The sense in which Carnap has here gone even further than the "logical idealism" of the Marburg school stands out especially clearly in §179 — entitled "The Task of Science." According to the Marburg school, as we have seen, the real individual object of empirical cognition is as a matter of fact never actually present in the methodological progress of science at all: this real empirical object remains always a never completed *X* toward which the methodological progress of science is converging. But Carnap, in §179, explicitly rejects this "genetic" view of knowledge:

> According to the conception of the Marburg school (cf. Natorp [*Grundlagen*] 18ff.), the object is the eternal X; its determination is an incompleteable task. In opposition to this it is to be noted that

> finitely many determinations suffice for the constitution of the object — and thus for its unambiguous description among the objects in general. Once such a description is set up the object is no longer an X, but rather something unambiguously determined — whose complete description then certainly still remains an incompleteable task.[60]

Carnap thus rejects the idea that the object of empirical knowledge, in contradistinction to the purely formal objects of mathematical knowledge, is to be conceived as a never-ending, necessarily incompleteable progression.[61] For Carnap, all objects whatsoever — whether formal or empirical, ideal or real — are rather to be defined or "constituted" at *definite finite ranks* within the hierarchy of logical types: there are no objects in the constitutional system that remain necessarily incomplete.[62]

In this sense Carnap completes the "logicization" of experience that the Marburg school had begun and, at the same time, arrives at an even more radical transformation of the Marburg tradition. For in the constitutional system of the *Aufbau,* epistemology is transformed into a logical-mathematical constructive project: the purely formal project of actually writing down the required structural definite descriptions within the logic of *Principia Mathematica.* This purely formal exercise is to serve, in particular, as a *replacement* for traditional epistemology in which we represent the "neutral basis" common to all traditional epistemological schools. All such schools are in agreement, according to Carnap, that

> cognition traces back finally to my experiences, which are set in relation, connected, and worked up; thus cognition can attain in a logical progress to the various structures of my consciousness, then to the physical objects, further with their help to the structures of consciousness of other subjects and thus to the heteropsychological, and through the mediation of the heteropsychological to the cultural objects. (Carnap 1928a, §178)

Since the constitutional system precisely represents this common ground of agreement within the neutral and uncontroversial domain of formal logic itself, all "metaphysical" disputes among the competing schools — disputes, for example, among "positivism," "realism," and "idealism" concerning which constituted structures are ultimately "real" — are thereby dissolved.[63] The fruitless disputes of the epistemological tradition are replaced by the seriousness and sobriety of the new mathematical logic, and philosophy (once again) becomes a science: for Carnap, a purely technical subject.[64]

By precisely representing some of the central ideas comprising the "logical idealism" of the Marburg school within the new mathematical logic of *Principia Mathematica,* Carnap has thereby injected the *neue Sachlichkeit* into philosophy itself. Philosophy becomes, in particular, an "objective" discipline capable (like the exact sciences) of cooperative progress and, in principle, of universal agreement as well; indeed, it has now become a branch of mathematical logic — the most "objective" and universal discipline of all. We have thus arrived at a conception of philosophy that, in Carnap's eyes, best serves the socialist, internationalist, and anti-individualistic aims of that cultural and political movement with which he most closely identifies.[65] And this "objectivist" and universalist conception of philosophy (based on the new mathematical logic) of course stands in the most extreme contrast with the particularist, existential-historical conception of philosophy we have seen Heidegger develop (based on an explicit rejection of the centrality of logic) — a conception that, in Heidegger's eyes, best serves the neoconservative and avowedly German nationalist cultural and political stance favored by the latter philosopher.[66] But what is of most interest, from our present point of view, is the extent to which both philosophers develop their radically new conceptions of philosophy by rigorously thinking through the ideas of late nineteenth- and early twentieth-century neo-Kantianism. By pushing these neo-Kantian ideas to their limits in two opposite directions, as it were, Carnap and Heidegger thereby contribute decisively toward defining and shaping the contemporary opposition between "analytic" and "continental" philosophical traditions with which we began.

Notes

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1. For eyewitness accounts see the report of L. Englert in Schneeberger 1962, 1–6; Pos 1949; and T. Cassirer 1981 — relevant parts of which are reprinted in Schneeberger 1962, 7–9.

2. Heidegger 1929a. This work appears in Heidegger 1991 together with appendices containing Heidegger's notes for his Davos lectures and a protocol of the Cassirer-Heidegger debate prepared by O. Bollnow and J. Ritter. These materials are also found in the English translation of Heidegger 1929a.

3. I am indebted to Thomas Uebel for first calling my attention to the fact that Carnap attended the Cassirer-Heidegger lectures and debate at Davos. Carnap reports on the occasion in ASP 025-73-03, entries from 18 March through 5 April 1929.

4. Heinrich Gomperz, the son of the famous historian of Greek philosophy Theodore Gomperz, was a professor of philosophy at the University of Vienna and the author of *Weltanschauungslehre* (1905). Karl Bühler was an important psychologist and psycholinguist; he founded the Psychological Institute at the University of Vienna in 1922. For Carnap's participation see ASP 025-73-03, entries for 24 May and 14 June 1930.

5. The two versions of these lectures, including reports on the discussions, are ASP 110-07-21 and ASP 110-07-19, respectively. I am indebted to Brigitte Uhlemann of the University of Konstanz for providing me with transcriptions from Carnap's shorthand.

6. Heidegger 1929b, 14, 18 [107, 111–12]. Carnap quotes selections from these passages in Carnap 1932, 231–32 [71–72]. (Pagination of the English translations appears in brackets.)

7. Carnap 1963c, 874–75. On the following page Carnap continues: "I think, however, that our [antimetaphysical] principle excludes not only a great number of assertions in systems like those of Hegel and Heidegger, especially since the latter says explicitly that logic is not applicable to statements in metaphysics, but also in contemporary discussions, for example, those concerning the reality of space or of time." Compare the remarks in Carnap 1963a, 42–43: "It is encouraging to remember that philosophical thinking has made great progress in the course of two thousand years through the work of men like Aristotle, Leibniz, Hume, Kant, Dewey, Russell, and many others, who were basically thinking in a scientific way."

8. See ASP 025-73-03, entry for 26 May 1928: "In the evening with Waismann at Neurath's. I read the Preface to the 'Logischen Aufbau' aloud; Neurath is astonished and overjoyed at my open confession. He believes that it will affect young people very sympathetically. I say that I still want to ask Schlick whether it is too radical." (Schlick did indeed think it was too radical: see entry for 31 May.)

9. Carnap became attracted to the antimilitarist internationalism of the "socialist worker's movement" already during the Great War (Carnap 1963a, 9–10).

10. For a general cultural and political history of this orientation see Willett 1978. For a specific discussion of the relationship between the Vienna Circle and the Dessau Bauhaus see Galison 1990. Not all members of the Vienna Circle shared in this orientation, however. Schlick, in particular, was attracted neither to Marxism nor to anti-individualism more generally. Thus, for example, Feigl poignantly describes Schlick's reaction when he was presented with the manifesto (Hahn, Neurath, and Carnap 1929) — which calls for a new internationalist and collaborative form of philosophy and, in keeping with this spirit, is not even signed by its authors — on his return from Stanford in 1929: "Schlick was moved by our amicable intentions; but as I could tell from his facial expression, and from what he told me later, he was actually appalled and dismayed by the thought that we were propagating our views as a 'system' or 'movement.' He was deeply committed to an individualistic conception of philosophizing, and while he considered group discussion and mutual criticism to be greatly helpful and intellectually profitable, he believed that everyone should think creatively for himself. A 'movement,' like large scale meetings or conferences, was something he loathed" (Feigl 1969, 646).

11. The literature on Heidegger's own political involvement is now enormous. See, in particular, Ott 1988; Farias 1987; and Schneeberger 1962. Wolin 1991 is a very useful selection — including a translation of Heidegger's notorious *Die Selbstbehauptung der deutschen Universität,* delivered in celebration of the new Nazi regime when he assumed the rectorate at Freiburg in May 1933. A particularly interesting contribution, locating Heidegger's involvement in the context of that of the other German philosophers of the time, is Sluga 1993.

12. Carnap 1932e, 238 [77]. Compare also the preface to *Logical Syntax of Language,* dated May 1934: "In our 'Vienna Circle' and in many similarly oriented groups (in Poland, France, England, USA and *in isolated cases even in Germany*) the view has currently grown stronger and stronger that traditional metaphysical philosophy can make no claim to scientific status" (Carnap 1934b, iii [xiii]; emphasis added). For Carnap's view (from Prague) of the situation in Germany and Central Europe in 1934 see Carnap 1963a, 34: "With the beginning of the Hitler regime in Germany in 1933, the political atmosphere, even in Austria and Czechoslovakia, became more and more intolerable.... [T]he Nazi ideology spread more and more among the German-speaking population of the Sudeten region and therewith among the students of our university and even among some of the professors."

13. In ASP 025-73-03, entry for 5 July 1932, Carnap triumphantly records the fact that there were 250 people in the audience at Berlin.

14. Neurath 1932a (reprinted in Neurath 1981, 572–73). Orthmar Spann was an especially virulent Austrian-Catholic right-wing ideologue of the time.

15. See, e.g., Neurath's remarks (made in 1933) in Neurath 1981, 597n: "Here [in Austria] there is not an exclusive dominance by metaphysics as it is practiced by Rickert, Heidegger, and others — through which those of a new generation become well-known through *geisteswissenschaflicher Psychologie, geisteswissenschaftlicher Soziologie* and similar things." Compare Carnap's remarks on Neurath's commitment to physicalism, unified science, and Marxism in Carnap 1963a, 22–24.

16. Compare also the following retrospective remarks of Neurath's (made in 1936): "The strong metaphysical trends in Central Europe are probably the reason that within the Vienna Circle the antimetaphysical attitude became of central significance and was purposefully practiced — much more, for example, than would have been the case with the adherents of similar tendencies in the United States, among whom a particular, more neutral common-sense empiricism is very widespread and where metaphysics could not exert the influence that it did in Germany, say.... It is entirely understandable that a Frenchman is at first surprised when he hears how the adherents of the Vienna Circle distance themselves in sharp terms from 'philosophers' — he thinks perhaps of Descartes and Comte in this connection, the others however of Fichte and Heidegger" (Neurath 1981, 743).

17. As is well known, *Introduction to Metaphysics* depicts Germany as Europe's last hope for salvation from Russian communism, on the one side, and American technological democracy, on the other, and contains Heidegger's notorious remark about the "inner truth and greatness" of the National Socialist movement (Heidegger 1953, 152 [166]).

18. I am indebted to Kathleen Wright for first calling my attention to this passage. As Wright also first pointed out to me, the noted Heidegger scholar Otto Pöggeler comments on *Introduction to Metaphysics* as follows: "Heidegger had sufficient taste not to deliver a previous version of his lecture in which Carnap's emigration to America was put forth as confirmation of the convergence between Russian communism and the 'type of thinking in America'" (quoted from Wolin 1991, 218–19). Given that Carnap did not emigrate until December 1935, however, whereas Heidegger's lectures were held in the summer of that year, Heidegger cannot be here referring to Carnap's emigration. It is more likely, for example, that he is referring to Schlick's trip to Stanford in 1929, which is prominently mentioned in the foreword to Hahn, Neurath, and Carnap 1929. The remark about Russian communism, on the other hand, almost certainly refers to Neurath's activities.

19. The dissertation appeared as Carnap 1922. Carnap defends a modified version of the Kantian synthetic a priori according to which the topological — but not the metrical — properties of space are due to the form of our spatial intuition. Bauch was influenced not only by his teacher Rickert but also by the more scientifically oriented neo-Kantianism

of the Marburg school — as well as by his colleague at Jena, the logician Gottlob Frege (who also greatly influenced Carnap, of course). Some discussion of Bauch in relation to both Frege and Carnap can be found in Sluga 1980. Sluga (1993) explains the depth and centrality of Bauch's involvement with Nazism — in comparison with which Heidegger's own engagement somewhat pales. Curiously, Carnap himself never mentions Bauch's political involvement.

20. Carnap explains his neo-Kantian philosophical training in Carnap 1963a, 4, 11–12. Writings of Cassirer, Natorp, and Rickert (as well as Bauch) play an important role in the *Aufbau:* see Carnap 1928a, §§5, 12, 64, 65, 75, 162, 163, 179.

21. Some of the most important epistemological works of the two traditions are Cohen 1902; Natorp 1910; E. Cassirer 1910; and Rickert 1892. Rickert 1909 and Natorp 1912 are very useful summary presentations of the two traditions.

22. See Husserl 1900–1901. For Husserl and the neo-Kantians see, for example, Natorp 1912, 198; and Rickert 1909, 227. Husserl's notion of *Wesenserschauung* plays a central role in Carnap's conception of "intuitive space" in Carnap 1922. All of Heidegger's earliest works fall squarely within the antipsychologistic tradition of "pure logic" and, accordingly, are dominated by the thought of Rickert and Husserl: these include "Neure Forschungen über Logik" (1912), his doctoral dissertation *Die Lehre vom Urteil im Psychologismus* (1913–14), and his habilitation *Die Kategorien- und Bedeutungslehre des Duns Scotus* (1915–16) — all of which are reprinted in Heidegger 1978. There is, of course, a close relationship between this "pure logic" tradition and the work of Frege: indeed, as is well known, it was Frege's review of Husserl's earlier *Philosophie der Arithmetik* that inspired the antipsychologistic polemic of the *Logical Investigations.* It is interesting to note also that Heidegger comments very favorably on Frege's work in his "Neure Forschungen über Logik" (Heidegger 1978, 20).

23. This "genetic" view of the object of empirical knowledge is common to Cohen, Natorp, and Cassirer. It is articulated with particular force by Natorp (in the works cited in note 21 above). Cassirer's achievement, in *Substance and Function,* is to make the view precise by finally articulating a coherent conception of logic (as the theory of arbitrary relational structures) — something that had eluded both Cohen and Natorp. Thus, for example, whereas Cohen and Natorp self-consciously attempt to align logic more closely with mathematics, they still continue to make essential use of the traditional classification of the forms of judgment.

24. Natorp (1912) then replies to Rickert's criticism. E. Cassirer (1929, 406 [348]) brings out the issue between Rickert and the Marburg school here with particular clarity: "Rickert's proof-procedure, in so far as it is simply supposed to verify this proposition [that number is not derivable from identity and difference], could have been essentially simplified and sharpened if he had availed himself of the tools of the modern logical calculus, especially the calculus of relations. For *identity* and *difference* are, expressed in the language of this calculus, *symmetrical* relations; whereas for the construction of the number series, as for the concept of an ordered sequence in general, an *asymmetrical* relation is indispensable."

25. Here Lask cites, among others, the theories of Herbart, Bolzano, Husserl, Rickert, Meinong, and (Heinrich) Gomperz (see Lask 1912, 23–24). Note that what Lask calls "formal logic" coincides with what Rickert calls "transcendental logic."

26. Lask (1912, 55) writes: "The 'form' of judgment, concept, inference, etc. is a completely different thing from form in the sense of the category. One best distinguishes these two kinds of form as structural form and contentful form." Kant's metaphysical deduction, by contrast, rests entirely on the idea that "the same understanding, through precisely

the same action whereby it brought about the logical form of a judgment by means of analytic unity, also brings about, by means of synthetic unity, a transcendental content in its representations in virtue of which they are called pure concepts of the understanding" (A79/B195).

27. For a discussion of Lask's argument from the point of view of the Marburg school see E. Cassirer 1913, 6–14. I am indebted to Werner Sauer for first calling my attention to this essay and for emphasizing to me, in this connection especially, the crucially important differences between Cassirer and the Southwest school.

28. The reference is to Lotze 1874, §§316–20. See, for example, Heidegger 1978, 170.

29. See Heidegger 1978, 214–89. As Heidegger notes, his discussion here is based on Rickert 1911.

30. Heidegger 1978, 273. The realm of valid sense enjoys this primacy because *all* realms of existence as such (the natural, the metaphysical-theological, the mathematical, and the logical itself) become objects of our cognition only through the mediation of the logical (Heidegger 1978, 287).

31. Heidegger's judgment of the superiority of Lask over Rickert emerges already in his dissertation of 1913–14 (Heidegger 1978, 176–77n).

32. Heidegger spent the years 1923–28 as associate professor at Marburg, after serving as Husserl's assistant at Freiburg from 1916 through 1922. It was at Marburg, through his lectures, that Heidegger established a reputation as one of the most brilliant and exciting young philosophers in Germany even before the appearance of *Being and Time.* The Marburg lectures in question appear as Heidegger 1979 and Heidegger 1976, respectively. To be sure, Husserl's own overcoming of neo-Kantianism is by no means complete from Heidegger's point of view — a point to which we shall return below.

33. Husserl's discussion of truth as "identification" occurs in Husserl 1900–1901, vol. 2, §§36–39. The discussion of "categorial intuition" then follows in §§40–58. For Heidegger's assessment of the relationship between these ideas and the work of Lask — which Heidegger sees as together destroying once and for all the Kantian "mythology" of a synthesis of understanding and sensibility, form and matter — see Heidegger 1979, 63–90.

34. When Husserl speaks of the realm of the logical and asks after an "epistemology of the logical" he has in mind the entire realm of a priori "essences" accessible to *Wesensanalyse* and *Wesenserschauung* (which include, for example, the a priori "essences" of spatial phenomena studied by geometry, of color phenomena studied by the a priori "eidetic science" of color, and so on). The very special structures studied by formal logic properly so-called (subject and predicate, and so on) represent, as we have just seen, only a tiny fraction — the most abstract part — of this "essential" realm.

35. This idea of phenomenology as a pure or "transcendental" psychology becomes fully explicit only in Husserl 1911; it is developed in elaborate detail in Husserl 1913. In the first edition of volume 2 of the *Logical Investigations* Husserl had misleadingly characterized phenomenology as "descriptive psychology" — which, as he himself immediately recognized, concealed precisely the "transcendental" relation in which he intended phenomenology to stand to (empirical) psychology. See Husserl 1911, 318n [115–16n].

36. See Heidegger 1978, 341–411. Heidegger there links his conception of "subjective logic" with the problem of the application of the categories on p. 407: "If anywhere, then precisely in connection with the problem of the *application* of the categories — insofar as one admits this in general as a *possible* problem — the *merely* objective-logical treatment of the problem of the categories must be recognized as one-sided." The attached footnote then emphasizes the importance of Lask 1912. For Husserl himself, on the other hand, since he developed the idea of phenomenology entirely independent of the Kantian and

neo-Kantian traditions, this problem of the application of the categories was never a problem. Husserl's own problem was always rather the relationship between the logical and the psychological — a problem that need not involve the relationship in general between the abstract and the concrete.

37. For Heidegger's assessment of Dilthey's conception of the subject as "living person with an understanding of active history" in contrast to Husserl's more formal conception of the subject, see Heidegger 1979, 161–71. The influence of Dilthey is further exhibited in 1916 in Heidegger's preface to his habilitation, with its call for philosophy to become *weltanschaulich* — that is, engaged in the concrete historical events of the time (Heidegger 1978, 191; and cf. 205 n. 10). This call contrasts sharply with Husserl's own arguments (in Husserl 1911, 323–41 [122–47]) — contra Dilthey — that philosophy *as a science* must be eternally valid and thus essentially unhistorical. It seems clear, moreover, from the remarks on Emil Lask's "distant soldier's grave," that Heidegger's call for a *weltanschaulich* philosophy here is directly connected with his attitude toward the Great War.

38. The idea of "being-in-the-world," together with the idea that the theoretical orientation toward the "present-at-hand" is founded on the more basic practical orientation toward the "ready-to-hand," is presented in Heidegger 1927, §§12–13 and is then developed in detail in the remainder of division 1.

39. For "Care as the Being of Dasein," see Heidegger 1927, §§39–44; for "Temporality as the Ontological Meaning of Care," see §§61–66; for "Temporality and Historicality" see §§72–77. In developing this conception of the essential "historicality [*Geschichtlichkeit*] of Dasein," Heidegger is, as emphasized in note 36 above, self-consciously following the work of Dilthey — work that he explicitly opposes to the "superficial" and "merely methodological" attempt to distinguish the *Geisteswissenschaften* from the *Naturwissenschaften* (based on the distinction between "generalizing" and "individuating" modes of concept formation) developed within the school of Windelband and Rickert. See the comments on Rickert in Heidegger 1927, §72, and compare the polemic called "Die Trivialisierung der Diltheyschen Fragestellung durch Windelband und Rickert" presented in Heidegger 1979, 20–21.

40. The analysis of "being-toward-death" and the ensuing possibility of "authentic" existence are presented in Heidegger 1927, §§46–60 — an analysis that is intended to present the "being of Dasein" (which was presented only fragmentarily, as it were, in the preceding sections) for the first time as a unitary and unified whole.

41. For the priority of "existence" over "essence" in the analytic of Dasein, see Heidegger 1927, §9. For Heidegger's diagnosis of the failure of Husserlian phenomenology as resting on a neglect of the question of the *existence* of "pure consciousness," see Heidegger 1979, 148–57, in particular 152: "Above all, however, this conception of ideation [that is, *Wesenserschauung*] as abstraction from real individuation rests on the belief that the What of any being is to be determined in abstraction from its existence. If, however, there were beings *whose What is precisely to exist and nothing but to exist,* then this ideational mode of consideration with respect to such a being would be the most fundamental misunderstanding."

42. The footnote to Heidegger 1927, §44 (p. 218), refers us to the sections on truth and "categorial intuition" in volume 2 of the *Logical Investigations* discussed above, along with the work of Lask. Heidegger warns us against relying exclusively on the first volume of the *Logical Investigations,* which appears merely to represent the traditional theory of the proposition (in itself) derived from Bolzano.

43. See Heidegger 1927, §44 (p. 221): "Dasein *can* understand *itself* as understanding from the side of the 'world' and the other or from the side of its ownmost

possibility-for-being [*aus seinem eigensten Seinkönnen*]. The last-mentioned possibility means: Dasein discloses itself to itself in and as its ownmost possibility-for-being. This *authentic* disclosedness shows the phenomenon of the most original truth in the mode of its authenticity. The most original and authentic disclosedness in which Dasein as possibility-for-being can be, is the *truth of existence.*"

44. The temporality of "authentic" existence is articulated in Heidegger 1927, §§61–66, the temporality of everyday "inauthentic" existence in §§67–71, the temporality of "historicality" in §§72–77. How the temporality of Dasein is actually the prior ground of the "ordinary conception of time" (namely, the all-embracing public time within which events are dated and ordered) is explained in §§78–81.

45. See Heidegger 1927, §33, "Die Aussage als abkünftiger Modus der Auslegung," and compare the discussion of Lotze's theory of "validity" in Heidegger's lecture course on logic from 1925–26 (Heidegger 1976, 62–68). This discussion clarifies the relationship Heidegger perceives between the "Cartesian" predicament of the world-less subject enclosed within its own contents of consciousness and the "Husserlian" predicament of the ideal subject isolated from all questions of real existence. In Husserlian terminology, it clarifies the relationship between the "phenomenological" reduction that withdraws our attention from the external world and focuses on the contents of consciousness themselves and the more radical "eidetic" reduction that then focuses only on the "essence" or formal structure of these conscious phenomena — arriving, in the end, at "pure" or "absolute" consciousness. Heidegger's idea is that if we once start with the "Cartesian" predicament, but nonetheless demand a kind of objectivity, then all we have left, as it were, is the contrast between change and constancy, the real and the ideal. We thus arrive at a conception of truth or objectivity on which truth is fundamentally necessary, "essential," or eternal truth; and, in this way, the denial of "naive realism" leads to "essentialism." And it is this last form of "essentialism" that is Heidegger's ultimate target.

46. Thus Heidegger 1927, §44 contains such provocative assertions as, "Before Newton's laws were uncovered they were not 'true' "; and "[these] laws became true through Newton, with him a being became accessible in itself for Dasein" (226–27). Given Heidegger's fundamentally historical conception of truth, together with his "existential conception of science" (§69, pp. 362–64), the meaning of these assertions is relatively straightforward: Newton arrived at the laws of motion by means of an "authentic projection" of a particular scientific framework in a given historical situation — the context of the scientific revolution of the sixteenth and seventeenth centuries (cf. §3, pp. 9–10, on "scientific revolutions"). Outside of this historical context, on the other hand, Newton's "discovery" and accompanying "assertion" of the laws of motion simply make no sense. For Heidegger, there is then no "valid sense" or "proposition in itself" beyond Newton's (and our) actual historical "assertions" capable of serving as a "vehicle" of "eternal truth." Nevertheless, *what* Newton discovered of course existed before Newton: "With the uncoveredness the being showed itself precisely as that being that was already there before. So to uncover is the mode of being of 'truth' " (p. 227).

47. See the remarks on the "objectivity [*Objektivität*]" of authentically historical truth in Heidegger 1927, §76, p. 395: "In no science are the 'universal validity' of standards and the pretensions to 'universality' that the They and its common sense require *less* possible criteria of 'truth' than in authentic history."

48. See n. 19 above.

49. Carnap 1963a, 20–22. An outline of Carnap's lecture to the Circle on 21 January 1925, bearing the title "Gedanken zum Kategorien Problem: Prolegomena zu einer Konstitutionstheorie," appears as ASP 081-05-03. The "Entwurf" manuscript has not yet been

found. A table of contents, bearing the dates 17 December 1924 and 28 January 1925 (a revision after the lecture in Vienna), appears as ASP 081-05-02.

50. Carnap 1928a, §75. The passage from Cassirer's *Substance and Function* to which Carnap is here referring (E. Cassirer 1910, chap. 4, §9) is a criticism of Rickert's well-known argument that concepts in the *Naturwissenschaften* cannot individuate (cf. note 39 above). Cassirer diagnoses Rickert's error here as stemming from a neglect of the essentially relational mode of concept formation of modern mathematics and logic. Carnap (1928a, §12) points out, again referring to this discussion of Cassirer's (and also to Rickert, Windelband, and Dilthey), that the "logic of individuality" desired in the *Geisteswissenschaften* can be attained precisely in the modern theory of relations.

51. For recent work on Kantian and neo-Kantian aspects of the *Aufbau,* see, for example Haack 1977; Moulines 1985; and Sauer 1985, 1989 (which particularly stress the importance of Cassirer and the passage from Carnap 1928a, §75); Friedman 1987, 1992b, and Richardson 1992b (which also emphasizes the importance of Cassirer and the Marburg school); and Webb 1992. Two recent extended treatments of the development of logical positivism, Coffa 1991 and Proust 1986, are also worth consulting in this connection.

52. See Carnap 1928a, §§100, 143. Cf. the remarks in Carnap 1963a, 18: "The system [of the *Aufbau*] was intended to give, though not a description, still a rational reconstruction of the actual process of the formation of concepts."

53. See especially the discussion of "ascension forms [*Stufenformen*]" in Carnap 1928a, pt. 3.B. There are exactly two such "ascension forms": namely, class and relation extensions (§40). As Carnap (1963a, 11) explains, he first studied *Principia Mathematica* — whose type-theoretic conception of logic pervades the *Aufbau* — in 1919.

54. See Carnap 1928a, pt. 4.A, and cf. §§67–94. The procedure of "quasi-analysis" is a generalization (to nontransitive relations) of the "principle of abstraction" employed by Frege and Russell in the definition of a cardinal number (see §73).

55. See Carnap 1928a, pt. 4.B, for the constitution of the physical realm — including the qualitative realm of ordinary sense-perception (§§125–35) and then the quantitative realm of mathematical physics (§136). As Carnap makes clear in §136, the constitution of the latter realm is based on his earlier methodological studies (Carnap 1923, 1924).

56. See Carnap 1928a, pt. 4.C. According to Carnap, *only* the purely abstract world of physics — and not the qualitative world of commonsense perceptual experience — "provides the possibility of a unique, consistent intersubjectivization" (§136; cf. §133).

57. For the independence of the definition of the visual field, in particular, from all phenomenal qualities, see Carnap 1928a, §86. For the importance of purely structural definite descriptions, see pt. 2.A, especially §16: *"[E]very scientific statement can in principle be so transformed that it is only a structural statement.* But this transformation is not only possible, but required. For science wants to speak about the objective; however, everything that does not belong to structure but to the material, everything that is ostended concretely, is in the end subjective." "From the point of view of constitutional theory this state of affairs is to be expressed in the following way. The series of experiences is different for each subject. If we aim, in spite of this, at agreement in the names given for the objects [*Gebilde*] constituted on the basis of the experiences, then this cannot occur through reference to the completely diverging material but only through the formal indicators of the object-structures [*Gebildestrukturen*]." For a fuller discussion, as well as detailed arguments against an empiricist-phenomenalist interpretation of Carnap's motivations, see my articles cited in note 51 above.

58. Carnap 1928a, §177. Section 116 presents the actual constitution of *sensations* —

defined via a purely structural definite description containing only the basic relation as a nonlogical primitive.

59. Carnap's type-theoretic sequential construction therefore takes the place of the "general serial form" Cassirer sees as expressing the essence of empirical knowledge. Carnap agrees with Cassirer, however, that this kind of methodological sequence is the ultimate "datum" for epistemology and, in particular, that the contrast between "being" and "validity"—which, as we have seen, generates fundamental problems for the Southwest school—therefore has only a *relativized* meaning in the context of such a sequence (see Carnap 1928a, §42; and cf. E. Cassirer 1910, 412–13 [311]). In Friedman 1992b, §3, I mistakenly read §42 as a *criticism* of the Marburg school and, in general, failed to draw the crucial distinction between the Marburg view of this question and that of the Southwest school. I am indebted to Alan Richardson and Werner Sauer for rightly protesting against this assimilation (cf. nn. 27 and 51 above).

60. The reference is to Natorp 1910, chap. 1, §§4–6; cf. E. Cassirer 1910, chap 7, especially 418–19 [315]. I am indebted to Alison Laywine for emphasizing to me the importance of this aspect of Natorp's view in the present connection.

61. Cf. E. Cassirer 1910, 337 [254]: "In contrast to the mathematical concept, however, [in empirical science] the characteristic difference emerges that the construction [*Aufbau*], which within mathematics arrives at a fixed end, remains in principle *incompleteable* within experience."

62. It is worth noting, in this connection, that the well-known technical problems afflicting the constitution of the physical or external world in the *Aufbau* appear in fact to undermine Carnap's attempt to distinguish himself from the Marburg school here. It appears, in particular, that Carnap's rules for assigning colors to points of space-time ($\mathbf{R}^4$) never close off at a definite set (that is, a definite relation between space-time points and colors) located at a definite rank in the hierarchy of logical types: for this assignment is to be continually revised as we progress to higher and higher ranks (Carnap 1928a, §§135, 136, 144). And this means, from the point of view of Carnap's own constitutional system, that the Marburg doctrine of the never completed *X* appears after all to be fully correct—at least so far as physical (and hence all higher-level) objects are concerned.

63. See again Carnap 1928a, §178: *"[T]he so-called epistemological tendencies of realism, idealism, and phenomenalism agree within the domain of epistemology. Constitutional theory presents the neutral basis* [neutrale Fundament] *common to all. They first diverge in the domain of metaphysics and thus (if they are to be epistemological tendencies) only as the result of a transgression of their boundaries."* All other properly philosophical disputes are similarly dissolved. Thus, for example, both sides in the debate over the relationship between the *Geisteswissenschaften* and the *Naturwissenschaft* are correct: cultural objects are constructed out of heteropsychological objects and the latter, in turn, out of physical objects; in this sense the theses of physicalism and the unity of science are correct. On the other hand, however, cultural objects nonetheless belong to a distinct "object sphere" in the type-theoretic hierarchy; in this sense the thesis of the autonomy and independence of the cultural realm is equally correct. See §56 and also §§25, 29, 41, 151. Cf. nn. 15, 39, and 50 above.

64. See the (first edition) preface to Carnap 1928a: "The new type of philosophy has arisen in close contact with work in the special sciences, especially in mathematics and physics. This has the consequence that we strive to make the rigorous and responsible basic attitude of scientific researchers also the basic attitude of workers in philosophy, whereas the attitude of the old type of philosophers is more similar to a poetic [attitude].... [T]he individual no longer undertakes to arrive at an entire structure of philosophy by a [single] bold stroke. Instead, each works in his specific place within

the *single* total science." Cf. also the discussion in Carnap 1963a, 13, on the impact of reading Bertrand Russell's *Our Knowledge of the External World* in 1921: Carnap is most impressed by Russell's description of "the logical-analytic method of philosophy" — together with its accompanying call for a new "scientific" philosophical practice.

65. In this respect, Carnap's identification with the *neue Sachlichkeit* is even more radical than Neurath's. For Neurath, unlike Carnap, makes no attempt to turn philosophy itself into an "objective" — purely technical — discipline. See the remarks on Neurath in Carnap 1963a, 22–24, 51–52, and cf. Uebel 1996 for an illuminating discussion of the relationship between Neurath's philosophy and his politics. This significant difference between Carnap and Neurath seems to be missed in the otherwise quite useful discussion of the relationship between the Vienna Circle and the *neue Sachlichkeit* in Galison 1990, which generally ignores the important areas of disagreement between the two philosophers.

66. Heidegger himself is perfectly explicit about the connection between his political engagement and his philosophical conception of the necessary "historicality of Dasein" in a well-known conversation (in 1936) reported by Karl Löwith (Wolin 1991, 142). Curiously, this crucial connection seems to be missed in Bourdieu 1988, an otherwise very interesting study of the relationship between Heidegger's philosophy and German neoconservatism.